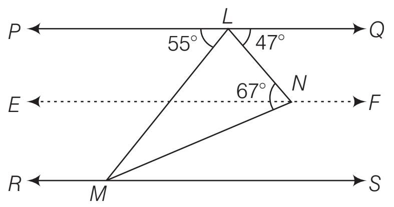
30. *(a)* Since, $PQ \parallel RS$

$\therefore \angle PLM = \angle LMS = 55°$ [alternate angle]

Draw a line EF which is parallel to PQ.

Then, $\angle QLN = \angle LNE = 47°$

$\because \angle ENL + \angle MNE = 67°$



$\Rightarrow 47° + \angle MNE = 67°$

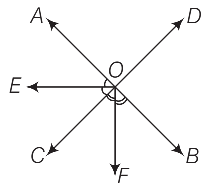
$\Rightarrow \angle MNE = 67° - 47°$

$\Rightarrow \angle MNE = 20°$

Similarly, $EF \parallel RS$, then

$\angle ENM = \angle NMS = 20°$ [alternate angle]

31. *(a)* Given, $\angle BOC = 130°$



$\because AOB$ is a straight line.

$\because \angle BOC + \angle AOC = 180°$ [linear pair]

$\Rightarrow 130° + \angle AOC = 180°$

$\Rightarrow \angle FOC + \angle FOC = 130°$

$\Rightarrow \angle AOC = 50°$

Now, $\angle BOC = 130°$

$\Rightarrow \angle BOF + \angle FOC = 130°$

[$\because OF$ is bisector of $\angle BOC$]

$\Rightarrow \angle FOC = 65°$

Now, $\angle AOC = 50°$

$\Rightarrow \angle AOE + \angle EOC = 50°$

$\Rightarrow \angle EOC + \angle EOC = 50°$

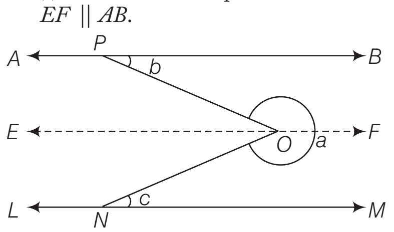
$\Rightarrow \angle EOC = 25°$

[$\because OE$ is bisector of $\angle AOC$]

$\therefore \angle EOF = \angle EOC + \angle FOC$

$= 25° + 65° = 90°$

32. *(c)* Draw a line parallel to AB i.e. $EF \parallel AB$.



$\therefore \angle EOP = \angle OPB = b$ [alternate angle]

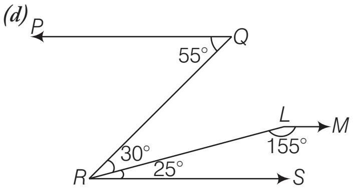
and $\angle EON = \angle ONM = c$ [alternate angle]

$\Rightarrow \angle PON = b + c$

$\because \angle PON + a = 2\pi$

$\therefore a = 2\pi - \angle PON = 2\pi - b - c$

33. *(d)*



$PQ \parallel RS$ ...(i)

Since, $\angle PQR = \angle QRS = 30° + 25° = 55°$ [alternate angle]

$PQ \parallel RS$ ...(i)

and $\angle SRL + \angle RLM = 180°$

$\Rightarrow RS \parallel LM$ ...(ii)

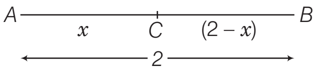
From Eqs. (i) and (ii), we get

$PQ \parallel LM$

So, the angle between the lines PQ and LM is 180°.

34. *(b)* Given, $AC^2 = AB \times CB$

$\Rightarrow x^2 = 2 \times (2 - x)$



$\Rightarrow x^2 = 4 - 2x$

$\Rightarrow x^2 + 2x - 4 = 0$

$\Rightarrow x = \dfrac{-2 \pm \sqrt{4 + 16}}{2 \times 1}$

$\Rightarrow x = -1 \pm \sqrt{5}$

Now, $BC = 2 - (-1 \pm \sqrt{5}) = 3 - \sqrt{5}$

(neglect $3 + \sqrt{5} \because 3 + \sqrt{5} > 2$)

35. *(c)* We have, $\angle DGE = \angle GCF = 95°$ [corresponding angles]

Also, $\angle GCF + \angle FCB = 180°$ [linear pair]

$\Rightarrow 95° + y = 180°$

$\Rightarrow y = 180° - 95° = 85°$

Again, $\angle ACB + y = 180°$

$\Rightarrow x = 180° - 85° = 95°$

Now, In ΔABC $\angle A + \angle B + \angle C = 180°$

$\Rightarrow 35° + z + 95° = 180°$

$\Rightarrow z + 130° = 180°$

$\Rightarrow z = 180° - 130° = 50°$

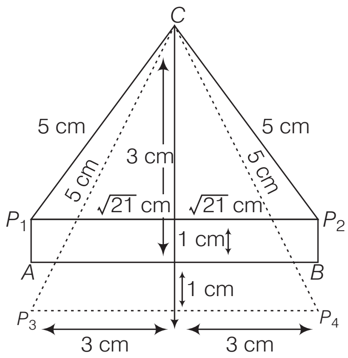
$\therefore x + y + z = 95° + 85° + 50° = 230°$

36. *(b)* Supplement of $z = 130°$

$\therefore \dfrac{1}{5}$th of $130° = \dfrac{130°}{5} = 26°$

37. *(d)* $\therefore$ Required number of points

$= 4\,(P_1, P_2, P_3, P_4)$



38. *(a)* Given that, $\dfrac{\alpha}{\beta} = \dfrac{1}{5}$

Let $\alpha = k$ and $\beta = 5k$

Sum of two complementary angles

$= 90°$

$\Rightarrow \alpha + \beta = 90°,$

$\alpha = 90° - \beta$

$\Rightarrow k = 90° - 5k$

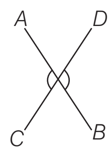
$\Rightarrow k = 15°$

$\therefore \alpha = 15°$ and $\beta = 75°$

$\therefore$ Difference between angles

$= 75° - 15° = 60°$

39. *(b)* Here, AB and CD are two lines.



If two straight lines intersect, then vertically opposite angles are equal.

40. *(b)* From the given figure,

$\angle LOQ + \angle QOP + \angle POM = 180°$ [straight line]

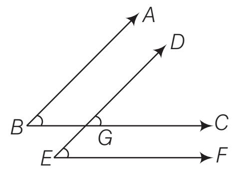
$\therefore (x° + 20°) + 50° + (x° - 10°) = 180°$

$\Rightarrow 2x° + 60° = 180°$

$\Rightarrow 2x° = 120°$

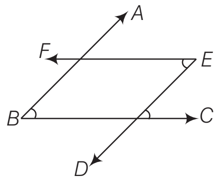
$\therefore x° = 60°$

41. *(b)* **Case I** When both pairs of arms are parallel same sense.



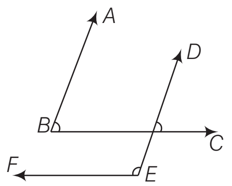
Here, $\angle ABC = \angle DEF$

Case II When both pairs of arms are parallel in opposite sense.



Here, $\angle ABC = \angle DEF$

Case III When one pair of arms is parallel in same direction and other pairs are parallel in opposite direction.



Here, $\angle ABC + \angle DEF = 180°$

So, the two angles are not equal but supplementary.

42. *(c)* Let the angle be θ.

$\therefore\ \theta = 80°$

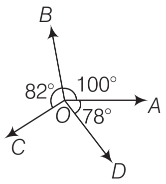
Complement angle

$$= 90° - \theta = 90° - 80° = 10°$$

[$\because$ sum of two complement angles is 90°]

$$= 10 \times \frac{\pi}{180°} = \frac{\pi}{18} \text{ radian.}$$

43. *(d)*



Given, $\angle AOB = \angle COD = 100°$,

$\angle BOC = 82°$

and $\angle AOD = 78°$

If, AOC is a straight line, then

$\angle AOB + \angle BOC = 180°$

$\Rightarrow\ 100° + 82° = 180°$

$\Rightarrow\ 182° \neq 180°$

If, BOD is a straight line, then

$\angle BOA + \angle AOD = 180°$

$\Rightarrow\ 100° + 78° = 180°$

$\Rightarrow\ 178° \neq 180°$

If, $\angle BOC$ and $\angle AOD$

are supplementary angles, then

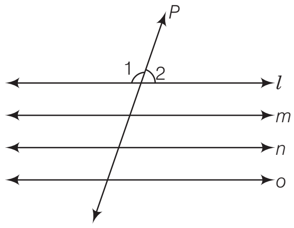
$\angle BOC + \angle AOD = 180°$

$\Rightarrow\ 82° + 78° = 180°$

$\Rightarrow\ 160° \neq 180°$

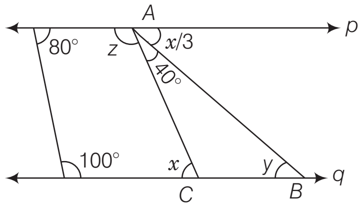
Hence, statements I and II are incorrect.

44. *(a)* If, line P intersect four parallel lines l, m, n and o, then 16 angles will be formed.



As these lines are parallel, hence distinct angle will be $\angle 1$ and $\angle 2$.

45. *(d)* In the given figure, lines p and q are parallel.



$\therefore\ x = 40° + \frac{x}{3}$ [alternate angles]

$\Rightarrow\ \frac{2x}{3} = 40° \Rightarrow x = 60°$

$y = x/3$ [alternate angle]

$y = \frac{60°}{3} = 20°,\ z + 40° + \frac{x}{3} = 180°$

$\Rightarrow\ z + 20° = 180° - 40°$

$\Rightarrow\ z = 120°$

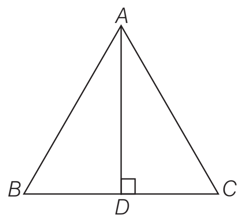
23

TRIANGLES

Generally (8-10) questions have been asked from this chapter. Generally questions are asked from the topic of pythagoras theorem, similarity of triangles and mid-point theorem.

TRIANGLE

A plane (closed) figure bounded by three line segments is called a triangle. It is denoted by Δ.



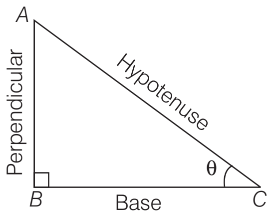
ΔABC has

- three vertices, namely A, B and C.
- three sides, namely AB, BC and CA.
- three angles, namely $\angle A$, $\angle B$ and $\angle C$.
- Sum of three angles of a triangle is $180°$. i.e., $\angle A + \angle B + \angle C = 180°$
- Area of a triangle $= \frac{1}{2} \times$ base $\times$ height $= \frac{1}{2} \times BC \times AD$

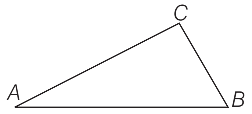
Classification of Triangle

1. On the Basis of Angles

(i) **Right Angled Triangle** A triangle in which one of the angle measures $90°$ is called a right angled triangle. The side opposite to the right angle is called its hypotenuse and the remaining two sides are called as perpendicular and base depending upon conditions. Here, ΔABC is a right angled triangle in which $\angle B = 90°$ and AC is hypotenuse.

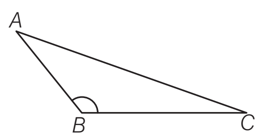


(ii) **Acute Angled Triangle** A triangle in which every angle measure more than $0°$ but less than $90°$ is called an acute angled triangle.
Here, ΔABC is an acute angled triangle.



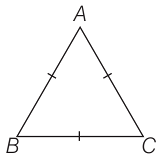
(iii) **Obtuse Angled Triangle** A triangle in which one of the angle measures more than $90°$ but less than $180°$ is called an obtuse angled triangle.

Here, $\Delta\ ABC$ is an obtuse angled triangle and $\angle ABC$ is the obtuse angle.



2. On the Basis of Sides

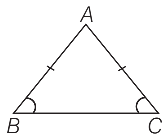
(i) **Equilateral Triangle** A triangle having all sides equal is called an equilateral triangle. Here ΔABC, is an equilateral triangle in which $AB = BC = AC$.



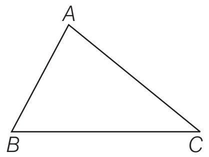
All angles are equal and are of measure 60°.

(ii) **Isosceles Triangle** A triangle in which two sides are equal, is called an isosceles triangle. Here, ΔABC is an isosceles triangle as $AB = AC$.

- Angle opposite to equal sides are equal.
  i.e. $\angle B = \angle C$



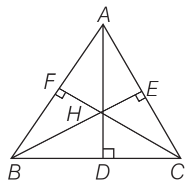
(iii) **Scalene Triangle** A triangle in which all the sides are of different lengths is called a scalene triangle. $\Delta\, ABC$ is a scalene triangle as $AB \neq BC \neq AC$.



Note The sum of the lengths of three sides of a triangle is called its perimeter.
So, perimeter of $\Delta ABC = AB + BC + AC$

Some Term Related to a Triangle

Altitudes and Orthocentre The altitude of a triangle is a line segment perpendicular drawn from vertex to the side opposite to it. The side on which the perpendicular is being drawn is called its base.

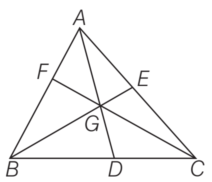


Here, In ΔABC, AD, BE and CF are altitudes drawn on BC, AC and AB, respectively.

- Altitudes of a triangle are concurrent.
- The point of intersection of all the three altitudes of a triangle is called its orthocentre.

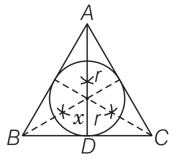
Medians and Centroid A line segment joining the mid-point of a side to the opposite vertex is called median.

Here, In ΔABC, AD, BE and CF are medians.



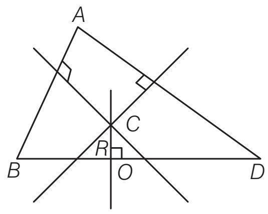
- The medians of a triangle are concurrent.
- The point of intersection of all the three medians of a triangle is called its centroid. It is denoted by G.
- Centroid divides the median in the ratio 2 : 1.
- A median bisects the area of the triangle i.e. $\text{Ar}(\Delta ABD) = \text{Ar}(\Delta ADC) = \frac{1}{2}\text{Ar}(\Delta ABC)$, etc.

Incentre and Angle Bisector The point of intersection of all the three angle bisectors of a triangle is called its incentre. It is denoted by I.



- The circle with centre I and touches all the sides is called incircle and radius of this circle is called inradius denoted by 'r'.
- Inradius $= \frac{1}{3} \times$ Height $= \frac{\text{Side}}{2\sqrt{3}}$,
  $ID \to$ Inradius, $AD \to$ Height

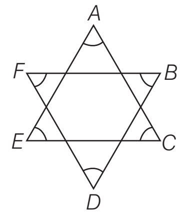
Perpendicular Bisector and Circumcentre The point of intersection of the perpendicular bisectors of the sides of a triangle is called its circumcentre. It is denoted by C.



The circle with centre C and passing through vertices A, B and C is called circumcircle. Radius of circumcircle is called circumradius denoted by R.

Circumradius $(CO) = \frac{2}{3} \times$ Height $= \frac{\text{Side}}{3}$

EXAMPLE 1. In the diagram given below what is the sum of all the angles $\angle A$, $\angle B$, $\angle C$, $\angle D$, $\angle E$ and $\angle F$?



a. 120° b. 180°
c. 290° d. 360°

Sol. **d.** Since, sum of the angles of a triangle is 180°.

In $\Delta\, AEC$, $\angle A + \angle C + \angle E = 180°$ ...(i)

and In $\Delta\, BDF$, $\angle B + \angle D + \angle F = 180°$ ...(ii)

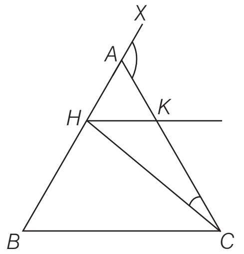
On adding the Eqs. (i) and (ii), we get

$$\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 180° + 180° = 360°$$

EXAMPLE 2. ABC is an isosceles triangle in which $AB = AC$, $CH = CB$ and HK is parallel to BC. If the exterior $\angle CAX = 137°$, then what is the measure of $\angle HCK$?

a. $68\frac{1}{2}°$ b. 43°
c. $25\frac{1}{2}°$ d. 137°

Sol. c. $\because\ \angle CAX = 137^\circ$



$\therefore \quad \angle ABC = \frac{1}{2}(137^\circ) = 68\frac{1}{2}^\circ$

$\therefore$ Again, $\quad BC = CH$ and $\angle ABC = 68\frac{1}{2}^\circ$

Therefore, $\angle CHB = 68\frac{1}{2}^\circ$, Therefore, $\angle HCB = 43^\circ$

Hence, $\quad \angle HCK = 68\frac{1}{2}^\circ - 43^\circ = 25\frac{1}{2}^\circ$

Some Important Results of a Triangle

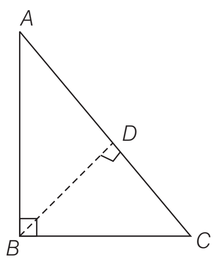
- If two angles of a triangle are equal, then the sides opposite to them are also equal.
- In a triangle, the side opposite to a greater angle is the longest side.
- The sum of all the three interior angles of a triangle is 180°.
- If a side of a triangle is produced, then the exterior angle so formed is equal to the sum of the two interior opposite angles.
- An exterior angle of a triangle is greater than either of the interior opposite angles.
- A triangle must have atleast two acute angles.
- If a perpendicular is drawn from the vertex of a right angled triangle to the hypotenuse, then the triangles on both sides of the perpendicular are similar to the original triangle and also to each other.
- In a right angle ΔABC, $\angle B = 90^\circ$ and AC is hypotenuse. The perpendicular BD is dropped on hypotenuse AC from right angle vertex B, then

(i) $BD = \dfrac{AB \times BC}{AC}$

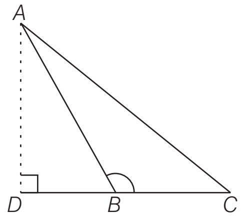
(ii) $AD = \dfrac{AB^2}{AC}$

(iii) $CD = \dfrac{BC^2}{AC}$

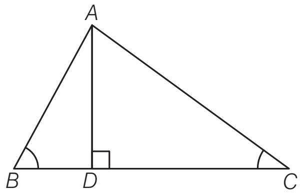
(iv) $\dfrac{1}{BD^2} = \dfrac{1}{AB^2} + \dfrac{1}{BC^2}$



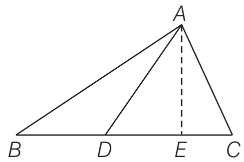
- The area of the equilateral triangle described on the side of a square is half the area of the equilateral triangle described on its diagonal.
- In ΔABC, if $\angle B > 90^\circ$ and AD is perpendicular dropped on BC, then $AC^2 = AB^2 + BC^2 + 2BC \cdot BD$



- In ΔABC, if $\angle B < 90^\circ$ and $AD \perp BC$, then $AC^2 = AB^2 + BC^2 - 2BC \cdot BD$



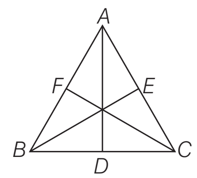
- In any triangle, the sum of the square of any two sides is equal to twice the square of half of the third side together with twice the square of the median which bisect the third side.



Here, AD is median, so

$$AB^2 + AC^2 = 2AD^2 + 2\left(\frac{1}{2}BC\right)^2$$

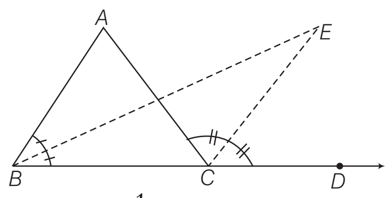
- In a $\Delta\, ABC$, three times the sum of the squares of the sides of a triangle is equal to four times the sum of the squares of the medians of the triangle.



So, in triangle, if AD, BE, FC are the medians, then

$$3(AB^2 + BC^2 + AC^2) = 4(AD^2 + BE^2 + CF^2)$$

- In an equilateral $\Delta\, ABC$, the side BC is trisected at D. Then, $\quad 9\,AD^2 = 7\,AC^2$
- When the bisector of one internal base angle and the bisector of external angle meet at a point then the formed angle is equal to one half of the vertical angle.



Here, $\quad \angle BEC = \frac{1}{2}\angle BAC$